

Pregame Pitcher Strikeout Projection - Resume Project Summary

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Companion to the full technical manuscript in `docs/paper/manuscript.md`.

Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and $P(K \geq L)$ for common prop lines L —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

Professional positioning

This project demonstrates a blended profile across:

- **AI/ML engineering:** end-to-end feature pipeline design, model freeze discipline, reproducible artifact/version controls, and production monitoring hooks.
- **Applied statistics:** chronological validation, bootstrap uncertainty intervals, calibration diagnostics, and explicit leakage audits.
- **Quant decisioning:** de-vig edge construction, CLV-based pre-registered decision gates, and policy-threshold governance under uncertainty.

Models and headline metrics

Component	Model	Role
Strikeout rate	Unweighted LightGBM (184 features)	Frozen production rate model
Batters faced (TBF)	Ridge (thin bullpen, 24 features)	Projected exposure
Counts / lines	Binomial / Poisson on projected TBF	Expected K and $P(K \geq L)$

Frozen rate model (2023–2024 chronological test): MAE / RMSE / $R^2 \approx$ **0.0787 / 0.0987 / 0.147**. Same test: Marcel-lite MAE $\approx$ **0.0826**; train-mean MAE $\approx$ **0.0854**.

Count layer (projected-TBF exposure): expected-K MAE / RMSE / $R^2 \approx$ **1.790 / 2.213 / 0.168**; line Briers roughly **0.12–0.22** (lines 3.5–7.5).

Full-stack evaluation

Evaluation gate	Result
Estimator tuning	Keep baseline LightGBM defaults; Ridge α tuned and persisted
Walk-forward stack backtest	Mean expected-K MAE $\approx$ 1.778 (3 expanding 2024 windows; $\sigma \approx 0.036$)
Calibration	Mean ECE $\approx$ 0.024 (11.C diagnose); production Platt post-hoc on <code>p_over_*</code> (chrono CV Δ ECE ≈ -0.008 ; raw retained). Fit on 2024 walk-forward OOS predictions; not yet refit on 2026 conditions.
Population policy	Metrics conditional on PA ≥ 9 ($\sim 3.5\%$ of first pitchers excluded)
Live market pilot (exploratory)	Paper-traded vs. real DK/FD lines with conservative policy gating. Pre-registered gate: $n \geq 100$ CLV samples + bootstrap CI excluding 0. Current status: inconclusive — sample is decision-scale ($n_{clv}=192$), mean CLV is positive ($\sim +0.54pp$), but CI still crosses 0 ($[-0.08pp, +1.16pp]$). Not a claim of market edge.

Resume-ready takeaway

Leakage-safe Statcast $\rightarrow$ rate x exposure stack with nested chronological selection. Clears a Marcel-lite talent floor on chronological test; opponent lineup is the only leave-family-out family with both-fold within-fold bootstrap support. Absolute R^2 remains limited (~ 0.15). A live CLV pilot against real sportsbook markets is running under pre-registered statistical gates and remains decision-inconclusive at manuscript freeze.

Portfolio claim: production-minded ML engineering under a strict information constraint, strengthened by statistical rigor and quant-style decision governance.

Full write-up: [docs/paper/manuscript.pdf](#)